Differential Equations: Slope fields and Solving Differential Equations — Problem Set

Warm-up

1. For the differential equation $\frac{dy}{dx} = \frac{x}{y}$, determine the slope at the point $(2, -1)$.

2. Solve the separable differential equation $\frac{dy}{dx} = 3x^2$ given $y(0) = 5$.

3. Find the general solution to $\frac{dy}{dx} = \cos x$.

4. Match the differential equation $\frac{dy}{dx} = y$ with its behavior: does the slope depend on x, y, or both?

5. State the value of $\frac{dy}{dx}$ at any point along the line $y = 1$ for the equation $\frac{dy}{dx} = (y-1)(x+2)$.

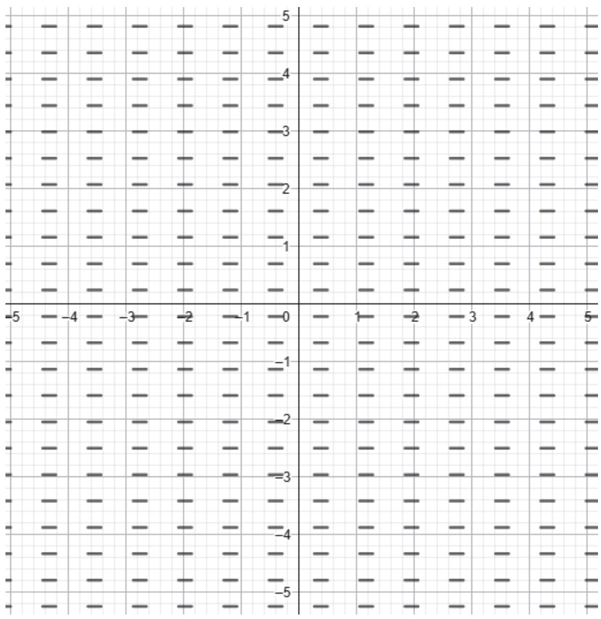
6. Solve $\frac{dy}{dx} = e^x$ with the initial condition $y(0) = 2$.

7. Separate the variables for the equation $\frac{dy}{dx} = xy$. (Do not solve).

8. If $\frac{dy}{dx} = f(x)$, describe the appearance of the slope field along any vertical line $x = c$.

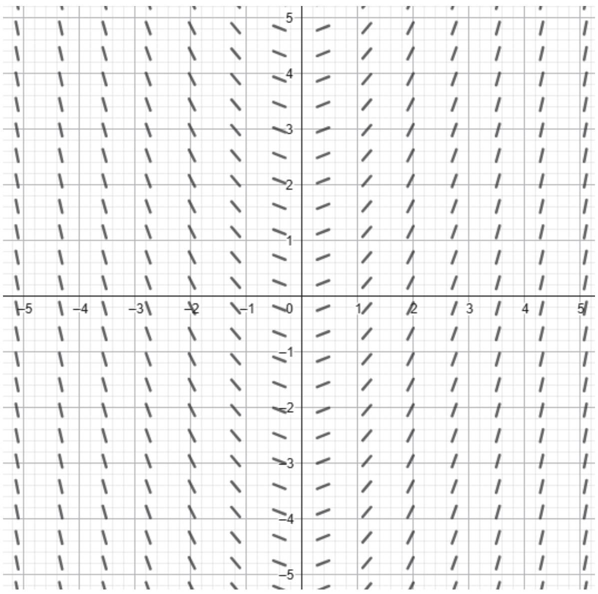
9. Given $\frac{dy}{dx} = \frac{1}{y}$, find the general solution in terms of y^2.

10. The direction field for a differential equation is shown below. Which of the differential equations below could give this direction field?



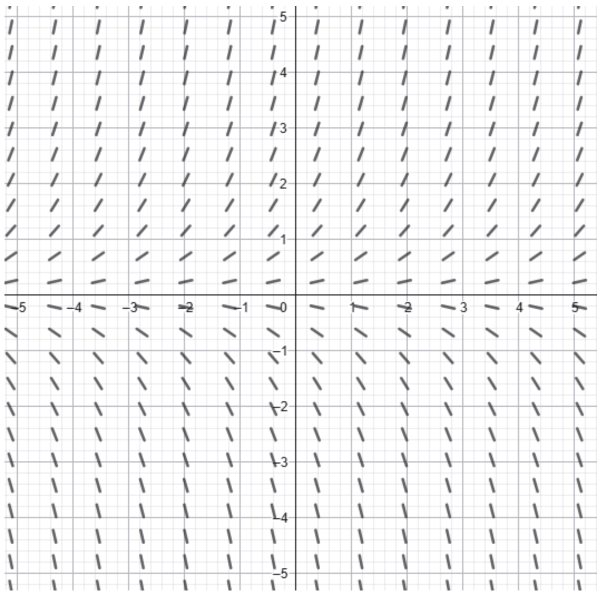
(A) $y' = 1$
(B) $y' = -1$
(C) $y' = y$
(D) $y' = 0$

11. The direction field for a differential equation is shown below. Which of the differential equations below could give this direction field?



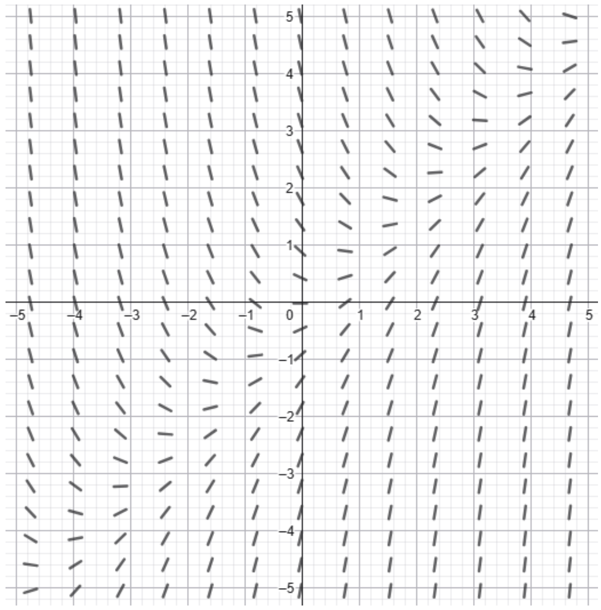
(A) $\frac{dy}{dx} = x$
(B) $\frac{dy}{dx} = -x$
(C) $\frac{dy}{dx} = y$
(D) $\frac{dy}{dx} = x - 1$

12. The direction field for a differential equation is shown below. Which of the differential equations below could give this direction field?



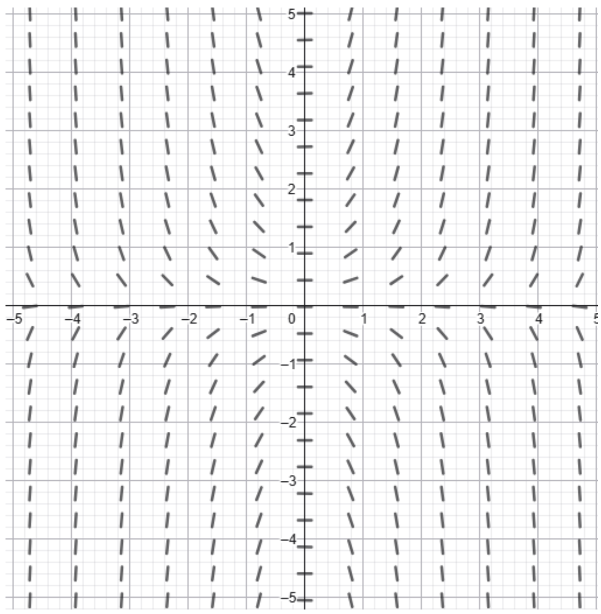
(A) $y' = \frac{1}{y}$
(B) $y' = y - 1$
(C) $y' = y$
(D) $y' = xy$

13. The direction field for a differential equation is shown below. Which of the differential equations below could give this direction field?



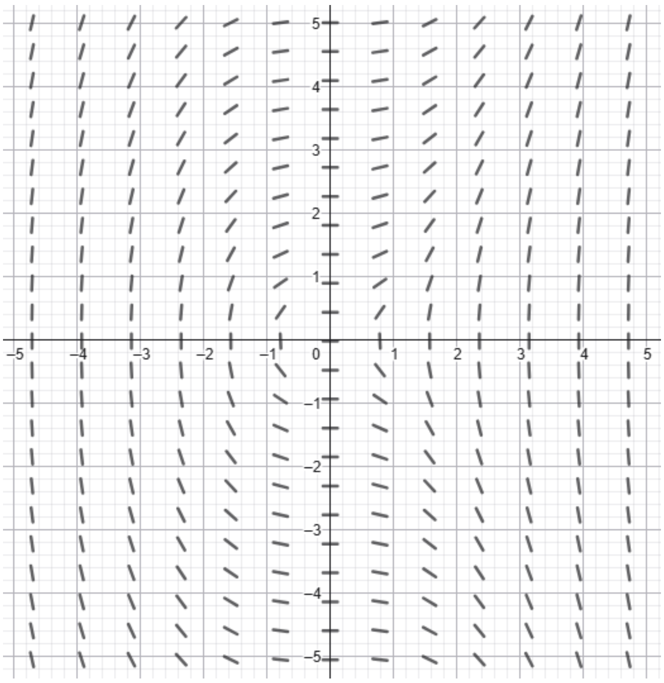
(A) $\frac{dy}{dx} = x - y$
(B) $\frac{dy}{dx} = x + y$
(C) $\frac{dy}{dx} = xy$
(D) $\frac{dy}{dx} = x + y - 3$

14. The direction field for a differential equation is shown below. Which of the differential equations below could give this direction field?



(A) $y' = x^2y^2$
(B) $y' = xy$
(C) $y' = \frac{x}{y}$
(D) $y' = \frac{y}{x}$

15. The direction field for a differential equation is shown below. Which of the differential equations below could give this direction field?



(A) $y' = \frac{y^2}{x}$
(B) $y' = \frac{x^2}{y}$
(C) $y' = \frac{x}{y}$
(D) $y' = \frac{x^2}{y^2}$

Skill-building

1. Solve the differential equation $\frac{dy}{dx} = \frac{x^2}{y^2}$ given the initial condition $y(0) = 1$.

2. Find the particular solution to $\frac{dy}{dx} = e^{x+y}$ passing through the origin $(0, 0)$.

3. Solve $\frac{dy}{dx} = \frac{y}{x}$ for $x > 0$ and $y > 0$, and express y as a function of x.

4. A slope field for $\frac{dy}{dx} = y - 2$ is drawn. Determine the equation of the particular solution that is a horizontal straight line.

5. Solve $\frac{dy}{dx} = y \cos x$ given that $y = e$ when $x = \frac{\pi}{2}$.

6. Find the general solution of $\frac{dy}{dx} = \frac{1+y}{1+x}$ by separating variables.

7. Given $\frac{dy}{dx} = xe^{-y}$ and $y(0) = 0$, find an expression for y in terms of x.

8. Consider $\frac{dy}{dx} = \frac{\sin x}{y}$. Find the particular solution that passes through $(\pi, 2)$.

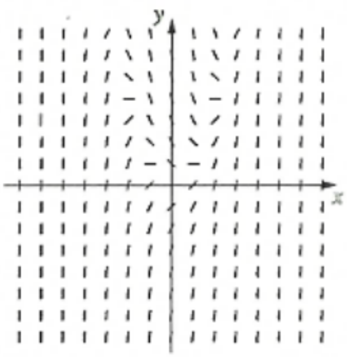
9. Solve the differential equation $\frac{dy}{dx} = \frac{e^y}{x^2}$ for y in terms of x.

10. For the differential equation $\frac{dy}{dx} = x - y$, find the equation of the isocline where the gradient is 1.

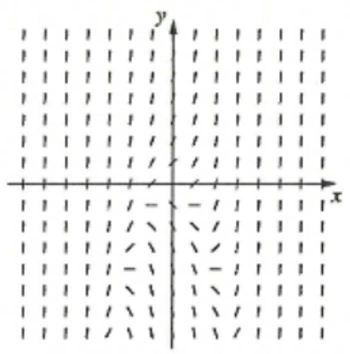
Easier Exam Questions

1. (Pymble 2025 Q10) Which of the following slope fields represents the differential equation $\frac{dy}{dx} = x^2 - y$?

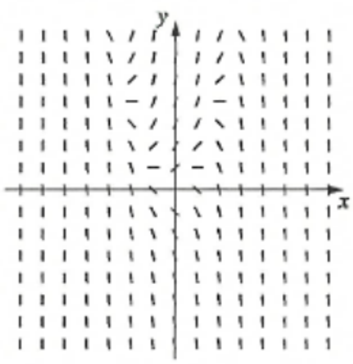
(A)



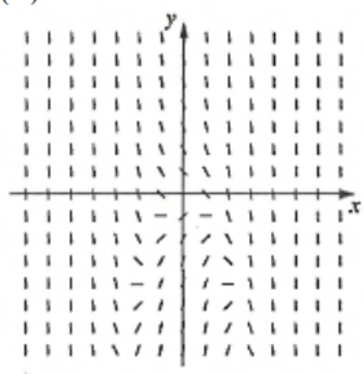
(B)



(C)



(D)



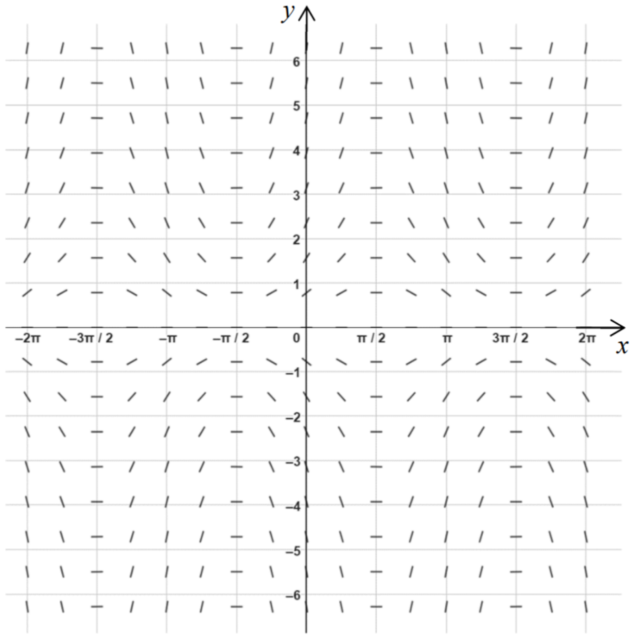
2. (Fort St 2023 Q7) Which differential equation is shown in the slope field below?

   (A) $y' = y\cos x$

   (B) $y' = y\sin x$

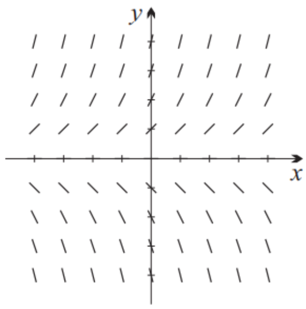
   (C) $y' = x\cos y$

   (D) $y' = x\sin y$

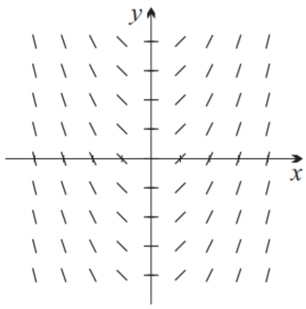


3. (Sydney Grammar 2024 Q6) Which of the following slope fields corresponds to the differential equation $y' = y - x$?

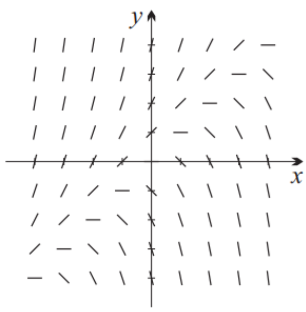
(A)



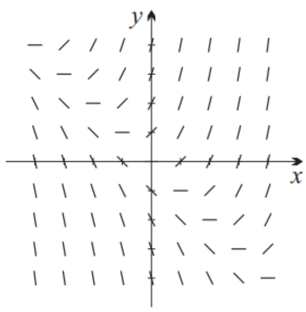
(B)



(C)



(D)



4. (Pymble 2025 Q14a) Given that $\frac{dy}{dx} = \frac{y-1}{x+2}$ and $y(6) = 3$, find the value of $y(1)$ **3**

5. (Penrith 2022 Q11b) Find the equation of the curve $r = r(\theta)$, which satisfies the differential equation **4**

$$\frac{dr}{d\theta} = \frac{r^2}{\theta}$$

With initial conditions $r(1) = 2$

6. (Sydney Boys 2024 Q11e) Consider the differential equation **3**

$$\frac{1}{y}\frac{dy}{dx} = \frac{\cos x}{1 - \sin x}$$

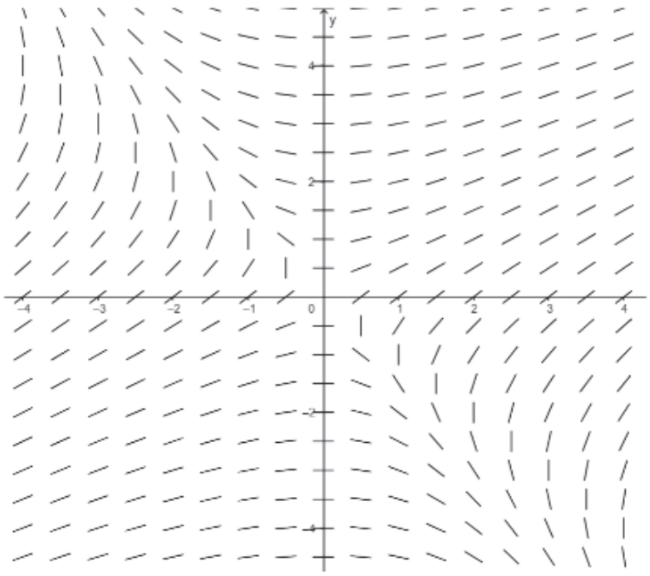
where $0 \leq x < \frac{\pi}{2}$ and $y > 0$
Given that $y = 1$ when $x = \frac{\pi}{6}$, express y as a function of x.

7. (Hunters Hill 2022 Q14b) Find the general solution of $\frac{dy}{dx} = e^{y-x}$ **2**

8. (Glenwood 2021 Q14a) Solve the differential equation $\frac{dy}{dx} = \frac{e^x}{2y}$, given the initial condition $y(2) = e$ **3**

9. (Trinity Grammar 2023 Q12b) The variables x and y are related by the differential equation $\frac{dy}{dx} = e^{2y}\cos 3x$. When $x = 0$, $y = 0$ Solve the differential equation, obtaining an expression for y in terms of x. **2**

10. (Merewther 2025 Q6) The slope field for the first order differential equation is shown below Which of the following could best represent the differential equation for the slope field?

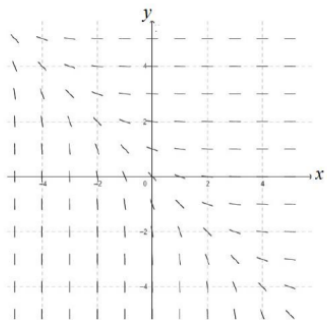


(A) $\frac{dy}{dx} = \frac{x+y}{x}$
(B) $\frac{dy}{dx} = \frac{x}{x+y}$
(C) $\frac{dy}{dx} = \frac{x}{y-2}$
(D) $\frac{dy}{dx} = \frac{y}{x-2}$

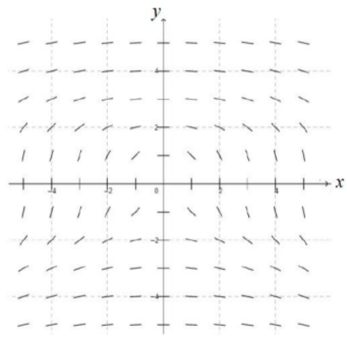
Harder Exam Questions

1. (Penrith 2023 Q7) Which of the following best represents the direction field for the differential equation $\frac{dy}{dx} = e^{x-y}$?

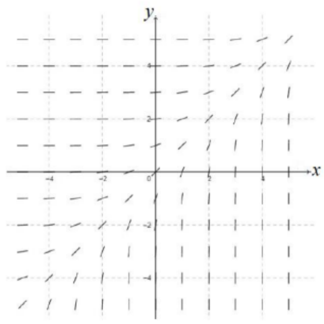
A.



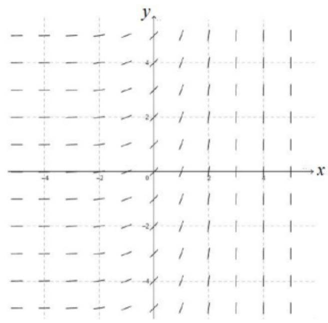
B.



C.

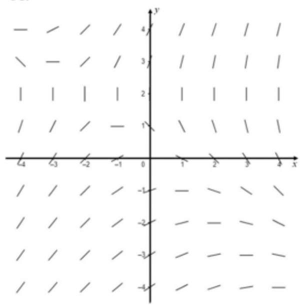


D.

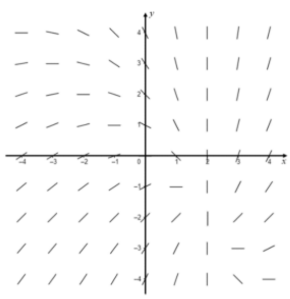


2. (Sydney Girls 2023 Q8) Which of the following slope fields represents the differential equation $\frac{dy}{dx} = \frac{x+y}{y-2}$?

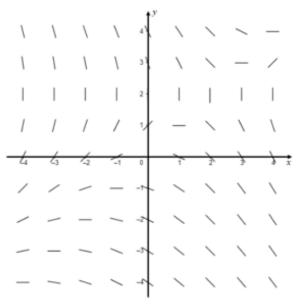
A.



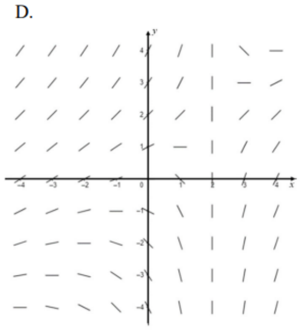
B.



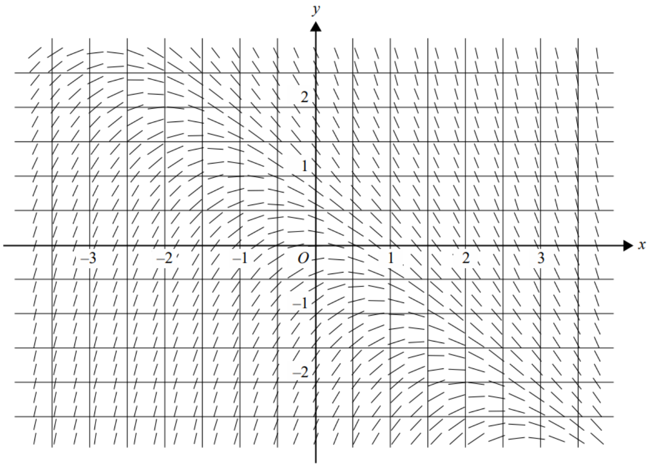
C.



D.



3. (Sydney Boys 2024 Q7)



The direction field for the differential equation $\frac{dy}{dx} + x + y = 0$ is shown above. A solution to this differential equation that includes (0, –1), would also include which of the following?

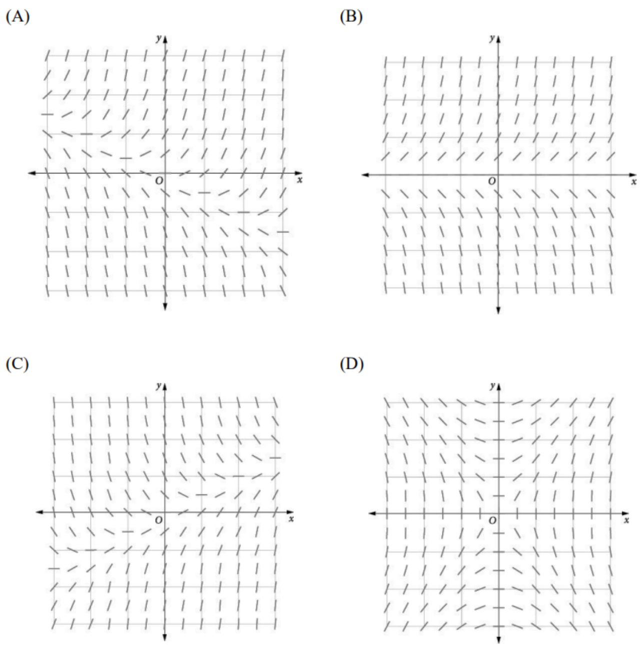
(A) $(3, -1)$

(B) $(3.5, -2.6)$

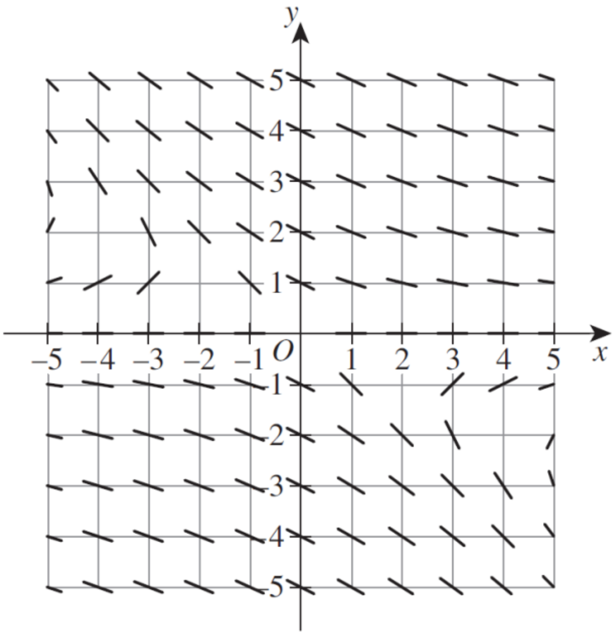
(C) $(-1.5, -2)$

(D) $(2.6, -1)$

4. (Sydney Girls 2021 Q10) Which of the following direction fields could have the differential equation $\frac{dy}{dx} = x - ky$ as a solution if $k > 0$?



5. (Abbotsleigh 2025 Q5) The direction field for a differential equation is shown below. Which of the differential equations below could give this direction field?



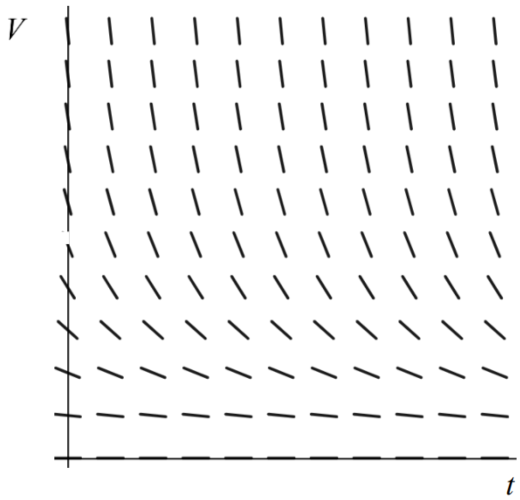
(A) $\frac{dy}{dx} = \frac{y}{2y+x}$

(B) $\frac{dy}{dx} = -\frac{y}{2y+x}$

(C) $\frac{dy}{dx} = \frac{y}{-2y+x}$

(D) $\frac{dy}{dx} = \frac{y}{2y-x}$

6. (Frensham 2021 Q8) A direction field for the volume of water, V megalitres, in a reservoir t years after 2021 is shown below.



According to this model, for $k > 0$, what is the value of $\frac{dV}{dt}$

(A) $-kV^2$

(B) $-\frac{k}{V}$

(C) $\frac{k}{V}$

(D) kV^2

7. (Fort Street 2023 Q3b) Find in the form $y = f(x)$ the solution of the differential equation $y' = \frac{2e^{-\frac{1}{2}y}}{\cos^2 x}$ given that $y = \ln 3$ when $x = \frac{\pi}{3}$ **3**

8. (Sydney Girls 2023 Q14e) Find the particular solution to the differential equation **2**

$$\frac{dy}{dx} = \frac{\cos x}{\cos y} e^{2\sin\left(\frac{x+y}{2}\right)\cos\left(\frac{x-y}{2}\right)}$$

which passes through the origin
(Note: In the expression above, e is raised to the power of $2\sin\left(\frac{x+y}{2}\right)\cos\left(\frac{x-y}{2}\right)$

9. (Abbotsleigh 2025 Q9) A first order differential equation is given by

$$\frac{dy}{dx} = 2y\ln y$$

Which of the following is a solution to the differential equation?

(A) $y = e^{2e^x}$
(B) $y = e^{e^{2x}}$
(C) $y = e^{e^{x^2}}$
(D) $y = 2e^{e^x}$

10. (Sydney Grammar 2021 Q12b) Solve the differential equation $\frac{dy}{dx} = 3x2e^{-y}$, given $y(0) = 1/$. Express your answer with y as the subject. **3**

11. (Glenwood 2022 Q12d) Solve the differential equation below, to give an equation for y in terms of x, given that $y(0) = 0$ **3**

$$\frac{dy}{dx} = e^x \cos^2 y$$

12. (Glenwood 2023 Q12c) Find the equation of the function which passes through the point $\left(1, \frac{\sqrt{6}}{3}\right)$, which satisfies the differential equation $\frac{dy}{dx} = \frac{x^3+1}{xy}$ **3**